

KALASH ABHAY KALA

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Technical Skills

Languages: Python, C, C++, SQL

Technologies/Frameworks: VERL, PyTorch, Hugging Face, LangGraph, Linux, GitHub, PowerBI,

Cloud Services: AWS (S3, Lambda, Sagemaker, SQS, Cloudwatch, RDS, Agentcore), Azure (OpenAI LLM, Document Intelligence)

Experience

Indian Institute of Science (IISc)

Bengaluru, Karnataka

Research Associate – AI/ML

January 2026 - Present

Research Focus: AI Safety, Uncertainty Estimation, and Multimodal Post-Training Alignment

- **Project: Multimodal Uncertainty Monotonicity Benchmark**
 - * **Developed a novel benchmark framework** evaluating multimodal LLM uncertainty along continuous, progressive ambiguity trajectories across spatial, scale, and audio dimensions.
 - * **Engineered naturalistic and synthetic visual/audio QA trajectories**, validated through a human study, resulting in a manuscript **recommended for acceptance at ACL Findings**.
- **Project: Internal State Mis-prediction & Risk Detection**
 - * **Pioneered an internal-state detection methodology** to isolate high-risk, highly confident model mis-predictions (high confidence incorrect and low confidence correct cases) across open-source LLMs.
 - * **Extracted and engineered features** from hidden states, attention maps, and MLP layers to train custom classification heads, boosting AUROC from **sub-50% to 77%**.
- **Project: Multimodal Abstention & Post-Training Alignment**
 - * **Architected a post-training pipeline using the verl framework** to train a multimodal LLM (Qwen-2-VL-7B) to safely abstain under structured visual ambiguity.
 - * **Synthesized a custom abstention dataset** by leveraging the Segment Anything Model (SAM) to programmatically perturb visual inputs based on targeted textual queries.

Autodesk

Bengaluru, Karnataka

Data Scientist

July 2024 - January 2026

- **Developed and deployed large scale web scraping and translation tool (WinSights)**, which scrapes and translates **50,000 websites/hour**
- **Engineered a serverless translation pipeline** using **AWS Lambda** and open-source models to replace AWS Translate, **cutting costs from \$60,000 to \$3,200** per 100,000 websites.
- **Scraped 1 million websites** which identified **\$1.2 million worth of sales opportunities at just \$35,000 of infrastructure cost**.
- **Developed and deployed on AWS** a real-time Slack listener using **GPT-4.1 LLM** that automates message filtering from Autodesk forums, increasing the product team's customer doubt resolution **rate by 300%**.
- **Fine-tuned and deployed a DistilBERT model on AWS** for a real-time Autodesk product recommendation engine, **processing 200,000 recommendations in under 40 minutes**

Data Science Intern

January 2024 - June 2024

- **Developed an Agentic Conversational chatbot** for the sales team to get relevant information about clients or raise a ticket if facing any technical issues. This **increased sales productivity and client outreach by 30%**
- **Created PowerBI Dashboard** to analyze usage of Autodesk Premier accounts credits, which includes; **Filtering, Paginated Grid, Pie Charts, Monthly and Annual Credit consumption Bar Charts, and Page Drill Through**

Projects

Auto Detection and Classification of Bleeding Areas in WCE Frames | Python, PyTorch

September 2023

- **Achieved 96% Accuracy** during classification of Bleeding and Non Bleeding Frames
- **Fine Tuned a Pre-trained Two Stage object detection algorithm called Faster R-CNN** for detecting Bleeding regions in the Bleeding Frames

Face Classification using Self Supervised Learning | Python, PyTorch

September 2023

- **Outperformed transfer learning** (using Efficient_net model) with much less labelled data
- Achieved Similar Accuracy as Supervised Learning in **half the number of epochs**
- Made Use of **Clustering as Pretext Task** for learning patterns from the unlabelled Face images

Education

Indian Institute of Information Technology, Vadodara

Bachelor of Technology in Computer Science, CPI: 8.7/10

December 2020 – May 2024

Gandhinagar, Gujarat

Achievements

- **Selected from a highly competitive pool (5% acceptance rate)** to present WinSights at Autodesk's premier internal technical conference (**TechX 2025**) in **Las Vegas**.
- **Selected for Amazon ML Summer School** top 3000 selects from India
- **Finalist** in Azadi Ka Amrit Mahotsav Hackathon, **conducted by the State Government of Gujarat**
- **First Prize** in Develop under 50 Hackathon, organised by IIIT Vadodara Technical Committee
- **First Prize** in HackIIITV Hackathon, organised by Coding Club of IIIT Vadodara
- **Secured Rank 960 / 20000** in LeetCode Weekly Challenge 341 ([LeetCode Profile](#))

Volunteering Service

Datakind Singapore Chapter

Data Science Volunteer

December 2022 – December 2023

- **Working on Container-based Action Routing Tool (CART)**. Used by Sanergy to provide routing service for sanitation workers in Kenya to optimise garbage collection, this is an **Open Source Tool**
- **Developed a python script to convert GeoJSON file to GPX format for showing the route on the app** created by Sanergy, made use of **Pandas** and additional open source library
- **Resolved the error caused by the recent update by debian in Dockerfile.dev**. Figured out that the error was caused due to the change of location of debian archives